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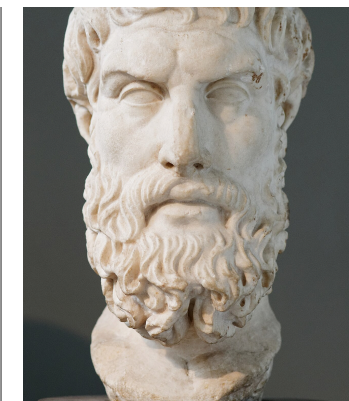
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Introduction: The induction problem

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Problem: What can we learn from (sometimes very few) examples?

You may have a look at the "[Sheep in Scotland](#)" [joke](#) to judge the difficulty of the task.



1–270 BC

le explanations:
ry is consistent with
p all theories.

ategy for

Solving the induction problem



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extends far beyond IQ tests.
It defines what is meant to understand
the situation.
It is also used to quantify the informational Content of the situation.
And last but not least,
it tells Artificial Intelligence how
to be... well,
intelligent.
Conclusion:
The best way to extrapolate from a
situation consists of Analyzing
that situation in a concise

Most IQ tests are based on induction. The problem is to find a relevant rule based on a few examples only.

IQ Test 1

0/1 point (ungraded)

What is the odd word out in this list?

cow, hen, pig, sheep.

This is a genuine induction problem: form a rule that covers most items (but happens to exclude one). Choose candidate rules that seem acceptable.

☒ Mammal. but excludes hen

Four legs, but excludes hen

Three letters in the word, but excludes sheep

Kosher or halal food, but excludes pig

Not "too" heavy, but excludes cow

✖

Explication
"Not 'too' heavy" is less acceptable than other rules (see next exercise).

Submit

You have used 1 of 1 attempt

 Answers are displayed within the problem

IQ test 2

1/1 point (ungraded)

Why may some solutions be less relevant than others in the previous exercise?

Less relevant rules are more complex because they require determining a complex quantitative value

Less relevant rules are more complex because they invoke new concepts

Less relevant rules are too simple because they are based on one single numerical value

Less relevant rules are too simple because they are based on one single concept

✔

Answer
Correct:
All rules invoke external concepts (legs, kosher, weight...). However, the rule invoking 'weight' is less relevant because it is **more complex**, due to the presence of a threshold value that requires a significant amount of information to be determined (here, about 500kg). The number of legs (4) remains simple in comparison.

Submit

You have used 1 of 1 attempt

✔ Correct (1/1 point)

If you are intested in tempting and yet false inductions, have a look at [Borwein integrals!](#)

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